

ME350 FA 2025 Section 002

Lab #7

Monday 12:30 PM - 2:30 PM

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Office Hours: 3:00 PM - 5:00 PM Monday (Zoom)



Agenda

- Reminders and Project Schedule
- Manufacturing Resources
- Mechatronics Lab Training
- Functional Testing

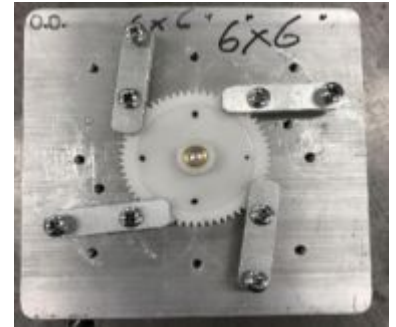


Reminders and Project Schedule

	Oct 5	Oct 6	Oct 7	Oct 8	Oct 9	Oct 10
Lecture			Lecture 12 - Transmission Gear Forces (MU)		Lecture 13 - Transmissions: Gears - Strength and Wear (MU)	
Homework						
Lab		Lab 6 - Gate Review 2 in Lab (X50 Assembly Room, 1510 GGB) Lab Assignment #5 Due - Transmission Selection Workbook Bill of Materials (BOM) Transmission CAD			Lab 7 - Functional Testing (X50 Assembly Room, 1510 GGB) Lab Assignment #6 Due - Manufactured GR1 Parts	Lab 7 - Functional Testing (X50 Assembly Room, 1510 GGB) Lab Assignment #6 Due - Manufactured GR1 Parts
Project Timeline		<-- Build & Assembly for Functional Testing				
	Oct 12	Oct 13	Oct 14	Oct 15	Oct 16	Oct 17
Lecture			No Lecture -- Fall Study Break		Lecture 14 - Intro to Mechatronics (JK)	
Homework					- Homework 4 Due (Transmissions 1) - Homework 5 Released	
Lab		No Lab -- Fall Study Break Lab 7 - Functional Testing TO BE COMPLETED ASYNCHRONOUSLY FOR MONDAY LABS Lab Assignment #6 Due - Manufactured GR1 Parts			Lab 8 - Sensors and Microcontrollers 1: Wiring, switches, and IR Prox sensor Lab Assignment #7 Due - Manufactured GR2 Parts	Lab 8 - Sensors and Microcontrollers 1: Wiring, switches, and IR Prox sensor Lab Assignment #7 Due - Manufactured GR2 Parts
Project Timeline						

Manufacturing Resources

- For modifications to shafts, gears, and motors, go to the [Manufacturing and Approval Packet References](#)
- Includes:
 - Procedure for spring-pinning a gear to a shaft
 - Custom machine shop fixtures for gears, shafts, motors, and gear & axles
 - Collet blocks (used to securely hold round parts in a mill vice)



Lab 8 (Mechatronics Lab)

- Full assembled mechanism (GR1 components, Transmission components, limit switch mount)
- You will need:
 - Safety Glasses
 - Your kit materials (we will have added electronic components)
- Watch the [Mechatronics Lab Training Video](#) **before Lab 8**
 - [Mechatronics Lab Resources](#)
 - [Mechatronics Safety and Overview Slides](#)



Parts for Assembling your Linkage

- Thrust washers are in the gray cabinet
- Shoulder bolts are in the gray cabinet
 - 10-32 shoulder bolts exist but are available in less quantities than the 10-24 bolts
- Steel washers are available in multiple locations
 - Some in X50 in the bins
 - Majority of washers available in machine shop
- Regular 10-24 bolts available in machine shop
- Lots of hardware available in the machine shop

Limit Switch Mounting

- Consider how you are going to attach the limit switch(s) to your linkage/baseplate.
 - It must contact a link (input or follower) at one end of the full range of motion. You can use two limit switches (one for each limit) to improve encoder accuracy.
 - The switch needs to compress before your linkage hits a hardstop (consider adjustability)
 - The [CAD is available here](#)



X50 Assembly Room 3D Printing

- Locally download the Bambu Lab Slicer, which is compatible with the X50 Assembly Room printers
 - Can be found here: <https://bambulab.com/en/download/studio>
- Download your part as a .STL File and upload to Bambu Lab
- Slice your part
- To obtain approval for your sliced file, either:
 - Attend office hours to review file and obtain official approval from a **GSI/IA**
 - Visit Kemal's office (1531A GGBL) and obtain official approval from Kemal
- Visit Kemal's Office (1531A GGBL) to check out a key to a printer anytime M-F from 8:30 AM - 4:30 PM. You will trade your MCard for the Key.
- Print on printers in X50 Assembly Room
- Return Key immediately after print completion or risk losing access to X50 Assembly room printers



Today, in order of priority:

- Assemble linkage
- Follow up on your drawings/MPs that haven't been approved
- Work on transmission components
 - If you're resizing input link holes in your drawing: write in the new hole size for your input shaft, ask Mike Klein/Don/Jon/Kemal to initial the update
 - If you're adding a new hole for a spring-pin, dowel-pin, or set-screw: create a new drawing showing dimensions on just the alteration
 - Start drawings/MPs for transmission parts

Keep Manufacturing! Full mechanism (including transmission components) should be assembled by Lab 8

Watch the Mechatronics Lab Training Video

Stay up to date with calendar on lecture Canvas Site!